

TEND ROBOTICS

Keeping elderly people safe in their own homes.

Pre-read for Raj

The thesis: wire ambient risk detection to a licensed clinician who can triage in seconds, on a path toward one remote nurse supporting one hundred homes.

Summary

Tend is a bet that one elder care nurse, given the right sensing and escalation stack, can safely watch over ten times the patients in their own homes.

Demand for elder care keeps growing while the workforce shrinks.¹ The cost of care is unsustainable for families and for the state. The system needs tools that multiply scarce clinical labour.

Detection wired to a live human already exists. Medical alert services and radar telecare route alarms to call handlers within seconds. The open position is narrower and stronger: a licensed clinician who can triage, assess and direct care, in private homes, at scale.

I would use the three months and the funds to test whether that position is real. That means at least 300 conversations with carers, providers, funders and families, a build-versus-buy call on the sensing layer, and signed paid-pilot intent from at least five firms.

This document names what would kill the idea. Frontier is where I run those tests and decide whether to commit the next decade.

I chose this problem deliberately, using a [framework](#) from Hardy Michel: a painful, worsening, structural problem, compounding insight from my physical AI and mechatronics background, and a mission that can hold a decade of my attention.

The problem

Somewhere tonight, an older person is on the floor of their own home, and no one knows yet. That gap, the hours between visits when nobody is watching, is the problem Tend exists to close.

Older people want to stay home, and the system cannot keep them there safely.

In the US, nearly nine in ten seniors want to age in place rather than enter a facility.²

The workforce to honour that wish does not exist.

Over 59 percent of US home care agencies report ongoing caregiver shortages.² HCAOA survey data puts the average agency's unstaffable work at 510 hours a month.³

CEDA projects Australia will be short 110,000 direct-care workers by 2030, rising past 400,000 by 2050.⁴

New Zealand's aged residential care holds the worst nursing vacancy rate in the health system: 15 percent in 2023.⁵

The cost of care is crushing, and families pick up the slack.

Residential care runs roughly NZD 60-90k a year in New Zealand,⁶ AUD 70-100k in Australia⁷ and USD 115-130k in the US.⁸ Means-tested subsidies shift part of this onto the state; the rest comes out of family estates.

In the US alone, AARP estimates 59 million family caregivers provided 49.5 billion hours of unpaid care in 2024, worth just over USD 1 trillion.⁹

And the quiet cost is loneliness.

Isolation is associated with roughly 50 percent higher dementia risk, 29 percent higher heart disease risk, and 32 percent higher stroke risk.¹⁰

Why physical AI will close this gap

Detection is a solved category. Contactless radar that spots falls in care homes and raises alarm ships today, and monitored response has existed for decades. Companionship, medication reminders and voice activated check ins are well-covered ground. The gap sits one layer up.

The open position is clinician-grade triage. No one wires ambient detection to a licensed clinician who can triage, assess and direct care in seconds, in private homes, on a path to one nurse per hundred homes. Newly FDA-cleared radar devices make contactless vital-sign monitoring possible with nothing touching the person.¹²

Remote clinical labour changes the economics. An offshore or centralised nurse acts as a triage and escalation specialist. Existing carers keep delivering hands-on visits; the nurse decides which alerts matter and what happens next.

Orchestration across a smart home. It's reasonable to expect household humanoids being deployed at scale in the next five years, and IoT devices are already commonplace. Tend will be the coordination layer that makes embodied AI deployable in a home care setting.

The moat is clinical and regulatory. The vetted clinician network, clinical governance, liability cover and payer contracts turn detection into a live clinical response..

The obvious objection, answered

The first question this idea earns: fall detection wired to a live responder has existed for decades at USD 30-50 a month, so what does Tend own?

Four things:

- **The responder.** Call handlers dispatch. A nurse assesses, resolves most alerts without an ambulance, and directs care on the ones that matter. False alarms stop becoming callouts and family panic. Slow deterioration gets clinical judgement.
- **The setting.** Incumbent radar deployments concentrate in residential care homes and telecare estates. Tend deploys into private homes: a different buyer, a different care setting, and thinner competition.
- **The clinical layer.** Detection vendors avoid it: clinical governance, triage protocols, liability cover and payer contracts. Vayyar sells detection and employs no nurses.
- **The orchestration.** One platform across sensors, companions and, later, humanoid bodies. No incumbent has shown ambition toward mobility or teleoperation.

Two parts of this answer are hypotheses, and I treat them as such. Whether buyers pay a clinician premium over dispatch-grade pricing is tested in the paid pilots (Appendix 2). Whether a remote nurse can lawfully triage across borders is an open licensing question: whether a protocol-bound triage role can sit below the registration line (Appendix 3), with an onshore fallback costed in Appendix 2.

The vision

Phase 1: sensing plus companion plus a live clinician

We orchestrate proven components rather than build them. Buying the sensing layer puts our whole effort into the layer no vendor touches: the clinical network and the live response. Existing carers keep visiting, now only when required. Our system watches the gaps and can alert a clinician live.

Phase 1 capabilities:

- Detect falls, including someone found on the ground between carer visits.
- Prompt medication at the right time and confirm it was taken, where memory is unreliable.
- Check that the person is eating and moving through their day.
- Track breathing rate, heart rate, and movement through contactless radar. No wearable, nothing touching the person.
- Watch daily patterns like room transitions, bathroom use, and time out of bed, and flag changes.
- Surface risk signals to a live clinician, who decides whether and how to act.

Candidate components: Vayyar for falls, presence and activity; Xandar Kardian or Origin AI for vital-sign radar; ElliQ or Andromeda's Abi for the companion layer. A build-versus-buy evaluation on the sensing layer is a month-one Frontier task (Appendix 1).

Phase 2: mobility and one nurse for one hundred homes

Once families trust a Tend device in the home, we add mobility and non-contact physical tasks.

We move into embodied robotics only once value is proven and the brand is trusted. We wait for bodies like Neo to be ready. These capabilities stay non-contact by design for now, which keeps us clear of medical-device regulation (Appendix 3).

Phase 2 capabilities:

- Move to the person and the incident. Bathrooms are a major risk area.
- Carry the nurse to the person for telepresence through teleoperation.
- Guide and lead: kitchen to a meal, door to a visitor, bed at night.
- Fetch and carry non-clinical objects around the home.
- Visually inspect the environment, then prompt and confirm at close range.

The destination is one remote nurse caring for one hundred homes.

As the technology keeps advancing, a robot will one day shower a person and cook them a meal. The tech is almost there. People are not ready yet, but trust is built slowly, and Tend will be the one they trust to do it.

Teleoperation is the bridge to autonomy. Early Phase 2 tasks run with a human in the loop, exactly as Neo ships today. When an incident happens, a nurse can take control through teleoperation and be present in seconds. As models improve, more of each task runs on its own, and one operator supervises more homes. The nurse-to-homes ratio that drives margin is a function of alert precision and task autonomy. Appendix 2 shows the arithmetic.

The Frontier plan

Months 1 and 2: understand deeply

- Run deep discovery with carers, providers, funders and families to pin down the exact moments where a live clinician creates value, and the ones where it does not. Target 300 conversations across the three months.
- Run the build-versus-buy evaluation on the sensing layer, with Vayyar, Xandar Kardian and Origin AI as candidates. Test data access, unit costs, and any exclusivities.
- Convert existing relationships across government health agencies, home-care providers and the healthcare community into pilot and evidence partners.
- Commit fully. Move to Tend full-time, with the IP held cleanly in the company and my full attention on the build.

Month 3: test demand

- Build a scrappy MVP and sign at least five firms as paid design partners, with pilot pricing agreed. Aged-care procurement will not close full contracts in a month; signed paid intent is the honest test of pull.
- Contract pilot triage capacity from an existing 24/7 telehealth nurse line rather than employing a roster, so the live-response promise is real from the first installed home (Appendix 2).

What would kill this

Frontier exists to kill weak ideas. These are the tests, with thresholds.

- **Nobody pays the premium.** If, after 300 conversations, no provider or funder will sign paid pilot intent at or above the placeholder band in Appendix 2, the clinician premium over dispatch-grade services fails.
- **Alerts are too noisy.** If field data from candidate sensors implies more than roughly ten clinician-minutes per home per day, one nurse per hundred homes is arithmetically out of reach, and the margin story fails.
- **Licensing blocks the ratio everywhere.** If counsel confirms remote triage into the home requires local registration in every target market, and onshore staffing cannot clear roughly one nurse per eighty homes at New Zealand rates, the long-run margin engine fails.
- **Homes refuse the sensing.** If a meaningful share of elders and families in discovery reject always-on sensing on privacy or dignity grounds, in-home deployment fails as designed.
- **No vendor opens the door.** If neither Vayyar nor an equivalent grants platform-level data access on workable terms, orchestration fails, and building radar hardware is the wrong company to build.

Any one of these ends the idea in its current form. I would rather learn that in month two than in year three.

Appendix 1: Competitors and components

The market splits into monitored response services, companion robots, monitoring software and home humanoids. None assembles the loop Tend assembles, though several own pieces of it.

Vayyar Care is the most direct challenge to our Phase 1 sensing claims, and our strongest candidate component. Vayyar ships contactless 4D imaging radar for fall detection today, at scale. Wall-mounted units detect falls automatically and monitor presence, time in bed, bathroom visits and activity patterns, without cameras or wearables.¹¹ Its commercial weight sits in residential care homes and telecare deployments rather than private homes. Alerts route to whoever the deployment defines: on-site staff in senior living, or 24/7 monitored response centres in telecare. Responders are trained call handlers. Vayyar does not employ nurses, hold clinical governance, or carry care liability.

Critically, Vayyar already partners. It runs an established partner programme and supplies radar to third-party telecare and monitoring platforms, so orchestrating its sensors asks nothing new of it. Open questions we test in month one: what data access a platform partner receives (processed alerts versus raw point cloud), unit and subscription costs at our scale, and whether exclusivities constrain a New Zealand or Australian deployment. The threat cuts the same way: a Vayyar monitoring partner could add clinician triage and close our gap. Speed into the clinical layer matters.

Monitored response is the incumbent category. US medical alert firms such as Bay Alarm Medical and Medical Guardian pair sensors with 24/7 professional monitoring. SafelyYou pairs fall detection with remote review in memory care. UK telecare routes radar detection to response centres within seconds.¹¹ This category proves demand for detection-to-response and sets the consumer price anchor near USD 30-50 a month. Its responders dispatch; they cannot clinically assess.

Intuition Robotics (ElliQ). A tabletop companion offering reminders, fall-risk assessment and a check-engine-light alert to clinicians. It has raised about USD 85 million.¹³ Washington authorised a statewide Medicaid reimbursement code for ElliQ in March 2026, the first social AI robot fully reimbursed by Medicaid, which proves the US payer channel for in-home care devices.¹⁴ It is expanding internationally, partnering with Kanematsu for a localised Japan launch in 2026.¹⁵ It is not mobile and does not support teleoperation. It flags a human rather than connecting one who responds live. Two cautions: ElliQ has its own clinician-alert ambitions and owns that Medicaid channel, so platform access is a question to test rather than assume. And for our sensing list, Vayyar is arguably the better-fitting component.

1X (Neo) is the leading in-home humanoid. It ships in 2026, teleoperated first, with elderly assistance named as a target use case. It brings the body without the care loop.¹⁶ We use this hardware if it stays ahead and switch makers if it does not. Staying body-agnostic keeps that choice ours.

Zuowei (Shenzhen Zuowei Technology) shows where China is going. It builds smart toileting and bathing robots and a broader eldercare platform, claiming China's first systematic range across toileting, bathing, eating, transfers, walking and dressing.¹⁷ This is the contact-care frontier we deliberately avoid for now. It marks where the market heads once trust and regulation catch up.

Andromeda Robotics (Abi) is a companion, and a possible partner. It places friendly companion robots in aged care, with no monitoring or teleoperation and heavy EU biometric exposure. We have a relationship with Andromeda and may partner to use their operating system if needed.

Sensi.AI monitors remotely, but audio only. It detects and alerts. It does not connect a clinician who intervenes.

CareClever (Cutii) is the cautionary tale. A capable mobile companion sold B2C at 90 euro a month. It failed on consumer subscription economics. Our read, which is inference rather than record, is that carrying detection promises without a response service compounded the problem. Either way the lesson holds: sell the response, and sell it to whoever funds care.

The gap, stated precisely: companionship, monitoring and dispatch are covered ground. No one wires ambient detection to a licensed clinician who can triage, assess and direct care in seconds, in private homes, on a path to one nurse per hundred homes. On current evidence, that position is open.

Appendix 2: Business model hypothesis and first-pass costing

Everything here is first-pass arithmetic. Validated costing is a core Frontier deliverable, built from the 300 conversations and real vendor terms.

Who pays?

- Providers, such as home care agencies looking to increase revenue by increasing the number of patients per employed nurse.
- Funders (e.g. Te Whatu Ora contracts, Australia's Support at Home, US Medicaid HCBS and Medicare Advantage). They buy to mitigate the strain on the health system caused by those ageing in place (keeping them out of hospital where possible) and to subsidise families' access to this technology,mitigating the economic pressure aging in place can put on families. ElliQ's statewide Washington Medicaid code shows the funder channel is real for in-home devices.¹⁴
- In New Zealand, how far this channel reaches depends on service classification. Personal care is funded at every income, while household help needs a Community Services Card. No category yet covers ambient monitoring with clinician response, so which test applies is a discovery question.

Price anchors. Dispatch-grade PERS sits near USD 30-50 a month. Residential care in New Zealand costs the equivalent of NZD 5,000-7,500 a month.⁶ A placeholder band for discovery: NZD 600-900 per home per month for sensing plus live clinical response. This number exists to be attacked in discovery, and it will move.

What a clinician seat costs. A response measured in seconds needs an always-staffed seat. Round-the-clock coverage takes roughly 4.5 nurse FTEs per seat once leave and sickness are covered. At a loaded New Zealand registered nurse cost of about NZD 140k,²⁴ one seat costs about NZD 630k a year. Clinician cost per home per month, before hardware and overhead:

Nurse-to-homes ratio	Clinician cost per home per month
1:40 (onshore, early)	~NZD 1,310
1:100 (target)	~NZD 525
1:250 (autonomy-assisted)	~NZD 210

What sets the ratio. Homes per seat equals productive clinician minutes divided by clinician minutes per home per day. One seat holds 1,440 minutes a day; assume half is usable after surge headroom. At two minutes per home per day, one seat covers several hundred homes. At fifteen minutes, fewer than fifty. False-alarm rates from field data decide which world we are in, which is why alert precision is kill criterion two.

The onshore fallback. If offshore triage is barred (Appendix 3), the model must clear margin onshore. At the placeholder price band, onshore economics need the ratio above roughly one nurse per eighty homes. That is the sensitivity the licensing work must resolve.

Pilot staffing within the SAFE. The \$250k SAFE does not fund a 24/7 nurse roster. For pilots we contract escalation capacity from an existing 24/7 telehealth nurse line, and we price pilots to cover that marginal triage cost. The roster gets built once the ratio maths and the funding channel are proven.

Appendix 3: Regulatory position

The claim this rests on

This pre-read says Tend is not a medical device. That claim is load-bearing, so it needs precision rather than absolutes.

Regulators do not classify a whole company. They classify each function by what it does and what it claims to do. Tend has two functions that sit on opposite sides of the line.

The sensing layer can stay outside device regulation. Contactless monitoring of activity, sleep and movement, framed as wellness and untied to a named disease, falls under the low-risk general wellness policy.¹⁸

The escalation function is the sensitive one. The moment software issues a clinical alert on its own, it looks like a medical device. Our design keeps a human making that call. We design each function to sit on the safe side of the line, and where a function cannot, we notify rather than seek approval.

Mission language versus product labelling. This document talks about keeping people safe and detecting falls, because that is the mission. Product labelling and marketing will be drafted to the wellness and decision-support line, and the safety promise attaches to the clinician service rather than to the software. Holding that discipline is a build constraint from day one.

United States: the wellness and decision-support line

The FDA reissued both governing guidance documents on 6 January 2026: General Wellness: Policy for Low Risk Devices, superseding the 2019 version, and Clinical Decision Support Software, superseding 2022.^{18, 19}

What stays clear of regulation. Since the 21st Century Cures Act, software that maintains or encourages a healthy lifestyle, unrelated to diagnosing or treating disease, is excluded from the device definition. The 2026 wellness guidance goes further: non-invasive sensors estimating physiological parameters such as heart rate can qualify as general wellness products, provided claims avoid disease references and diagnostic thresholds.¹⁸

Where the line is crossed. The same guidance is explicit that alerting for clinical purposes, alerts prompting clinical action, and disease-specific thresholds disqualify a product from wellness status.¹⁸ Tend's escalation would cross that line if the software made the clinical decision. Our design keeps it on the safe side in two ways. A licensed clinician interprets the signal and decides whether to act; the software only surfaces information. And the escalation logic stays transparent to that clinician. The 2026 CDS guidance keeps software outside the device definition when a professional can independently review the basis for a recommendation, and warns that black-box tools are more likely treated as devices.¹⁹

The honest caveat. This is a design discipline rather than a loophole. If we marketed the software as catching falls automatically, or let it escalate without a clinician, the exemption falls away. Our liability position and our regulatory position are the same discipline.

New Zealand: notification today, a new Act on the horizon

New Zealand is our beachhead, and its current regime is materially lighter than the US or EU.

There is no approval step today. Under the Medicines Act 1981 there is no pre-market approval system for medical devices. The only duty is notifying the WAND database within 30 days of taking a device to market, and notification is not an assessment of safety or performance.²⁰ Even if part of Tend's stack were treated as a low-risk device here, the obligation is to notify. That removes approval timelines that would otherwise gate a pilot.

The window has a clock on it. The Government is developing a Medical Products Bill to replace the Medicines Act, intended for introduction to Parliament in 2026 and taking effect around the end of the decade, with a longer transition for devices.²¹ We treat the light regime as a launch window rather than a permanent state. We hold safety documentation to the incoming standard from the start, because Medsafe can request it now and will require it later.

European Union: the reason we minimise biometrics

Competitors relying on facial recognition and emotion detection carry heavy EU exposure. The official basis is worth stating, because avoiding it is a deliberate design choice for us.

Emotion recognition is high-risk by default. The EU AI Act lists AI systems intended for emotion recognition among its high-risk category, which carries strict obligations.²² A narrow exception exists for systems designed for medical or safety purposes, which a safety-focused eldercare use may fit, though high-risk obligations still apply.²²

Biometric data is doubly protected. Under the GDPR, biometric data used to identify a person is a special category with heightened protection, layered on top of the AI Act.²³

Our design minimises biometric processing. We favour contactless movement and vital-sign sensing over facial recognition and emotion inference. That keeps us clear of the heaviest EU obligations our competitors invite.

The offshore clinician: an open question we flag honestly

The long-term model envisions a remote nurse, possibly offshore, triaging into many homes. This raises scope-of-practice and licensing questions that vary by country and are unsettled.

A clinician who assesses and directs care into a patient's home is generally practising in the patient's jurisdiction, which usually requires local registration. Triage and escalation may be treated differently from hands-on care, and the boundary is untested for this model.

We do not claim this is solved. The offshore ratio that drives long-run margin depends on regulatory clarity we will pursue with counsel. Appendix 2 shows the onshore fallback the model must survive if the answer is no.

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